

EC-350 AI and Decision Support Systems

Week 13 Artificial Neural Network - Perceptron

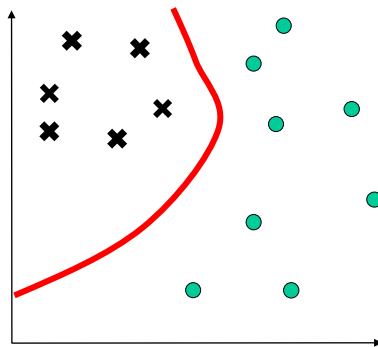
Dr. Arslan Shaukat



Acknowledgements: Lecture slides material from
Duda, Hart and Stork, Dr. Gavin Brown

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Reminder: A Decision Boundary



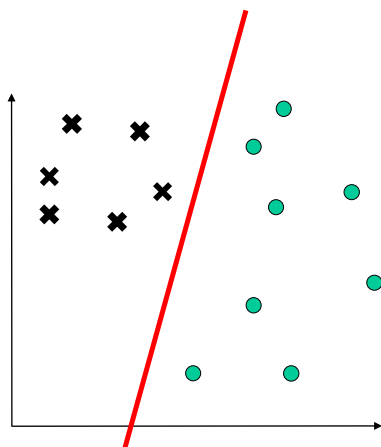
High accuracy

High space complexity

High time complexity

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A (Linear) Decision Boundary



Represented by:

*One artificial neuron
called a "Perceptron"*

Low accuracy (mostly)

Low space complexity

Low time complexity

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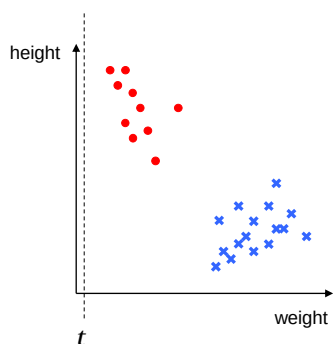
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A Simple Rule



if ($weight > t$) then "player" else "dancer"

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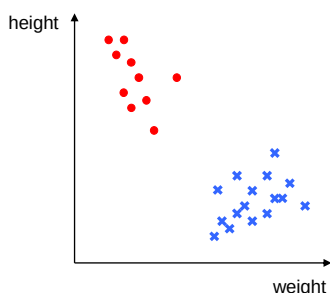
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Algorithm to Find a Good Threshold?

```
t=40
while ( numMistakes != 0 )
{
    t = t + 1
    numMistakes = testRule(t)
}
```

The threshold classifier (also known as a “Decision Stump”)



if ($weight > t$) then "player" else "dancer"

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Three “Ingredients” of a Machine Learning System

“Model”

The final product, the thing you have to package up and send to a customer. A piece of code with some parameters that have been set.

“Error function”

The performance criterion: the function you use to judge how well the parameters of the model are set.

“Learning algorithm”

The algorithm that optimises the model parameters, using the error function to judge how well it is doing.

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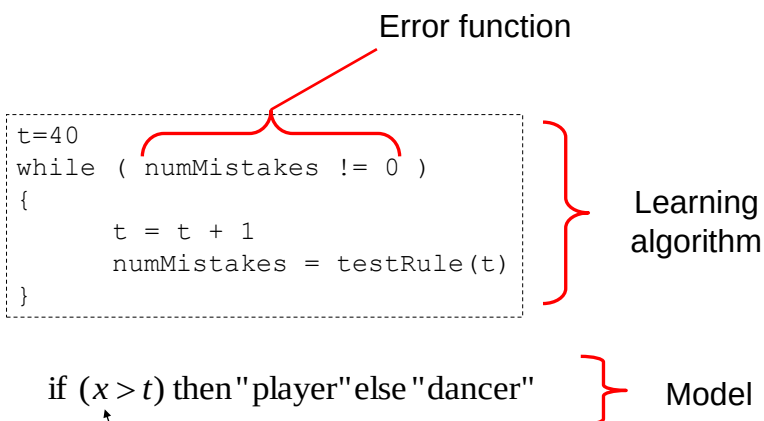
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Three “Ingredients” of a Threshold Classifier



General case – we're not just talking about the weight of a rugby player – a threshold can be put on any feature 'x'.

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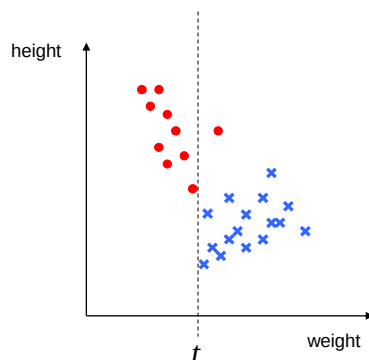
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Algorithm to Find a Good Threshold?



Our model does not match the problem!

if ($weight > t$) then "player" else "dancer"

1 mistake...

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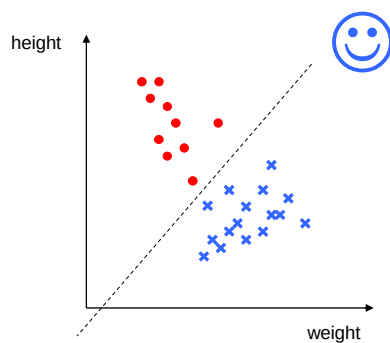
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Algorithm to Find a Good Threshold?



But our current model
cannot represent this...

if ($weight > t$) then "player" else "dancer"



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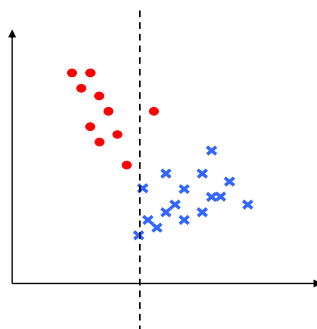
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We Need a More Sophisticated Model...

if ($x > t$) then "player" else "dancer"



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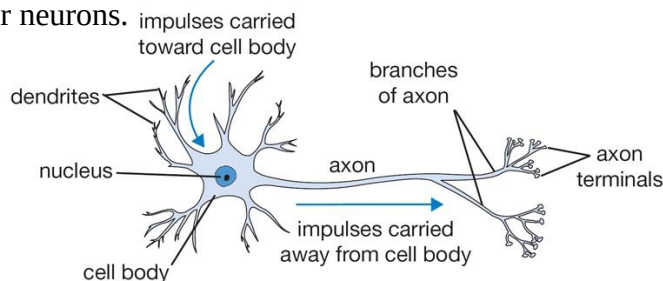
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Biological Motivation

- Basic computational unit of the brain is a **neuron**.
- Approximately 86 billion neurons can be found in the human nervous system and they are connected with approximately 10^{14} - 10^{15} **synapses**.
- Each neuron receives input signals from its **dendrites** and produces output signals along its (single) **axon**.
- The axon eventually branches out and connects via synapses to dendrites of other neurons.



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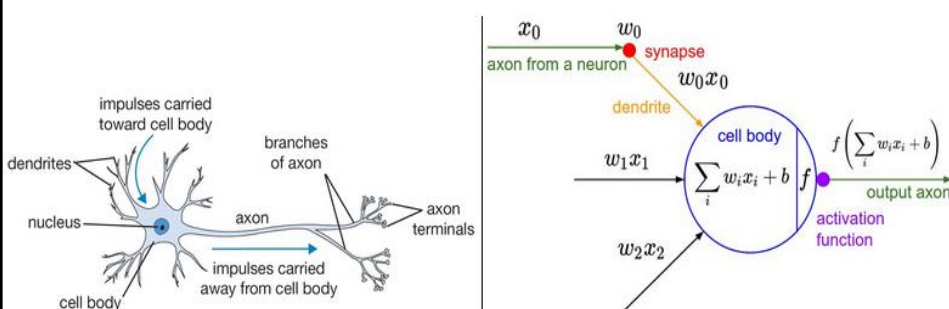
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From Biological Neuron to its Model



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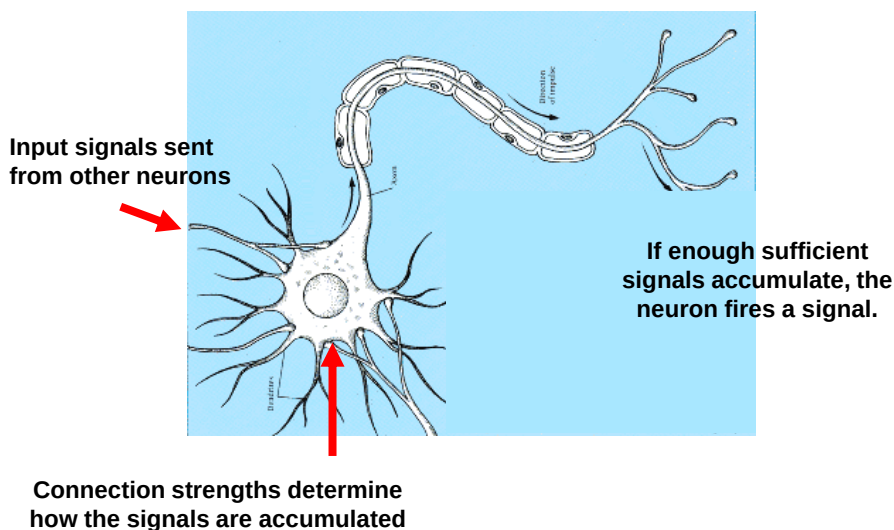
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Neural Network



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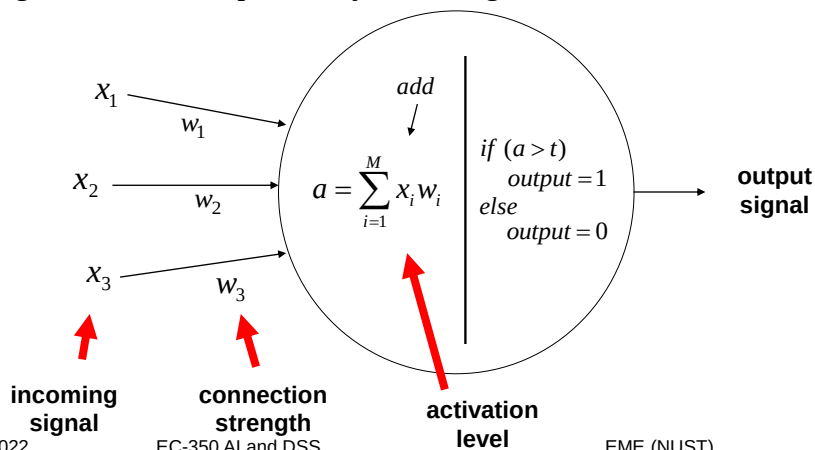
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Artificial Neural Network

- Input signals 'x' and weights 'w' are multiplied
- Weights correspond to connection strengths
- Signals are added up – if they are enough, FIRE!



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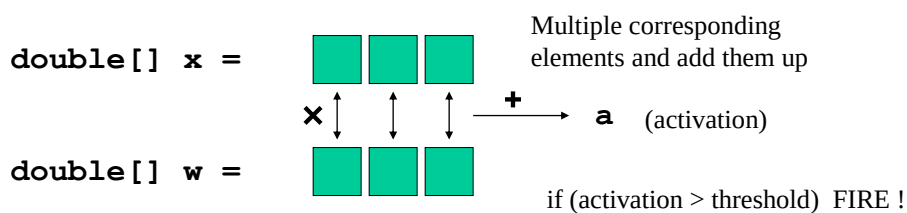
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ANN - Calculation

$$a = \sum_{i=1}^M x_i w_i$$

Sum notation

(just like a loop from 1 to M)



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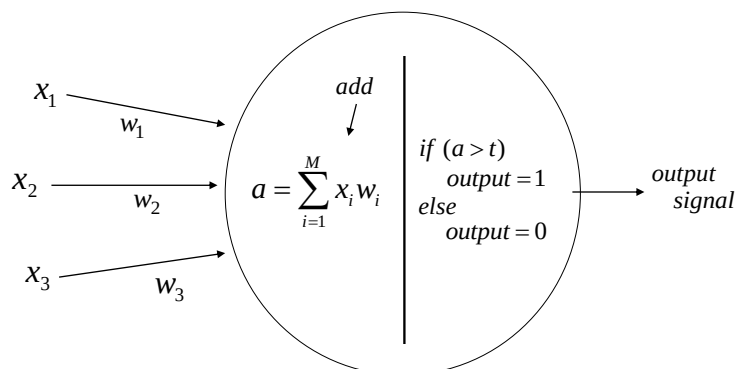
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Perceptron Decision Rule

$$\text{if } \left(\sum_{i=1}^M x_i w_i \right) > t \quad \text{then output} = 1, \text{ else output} = 0$$



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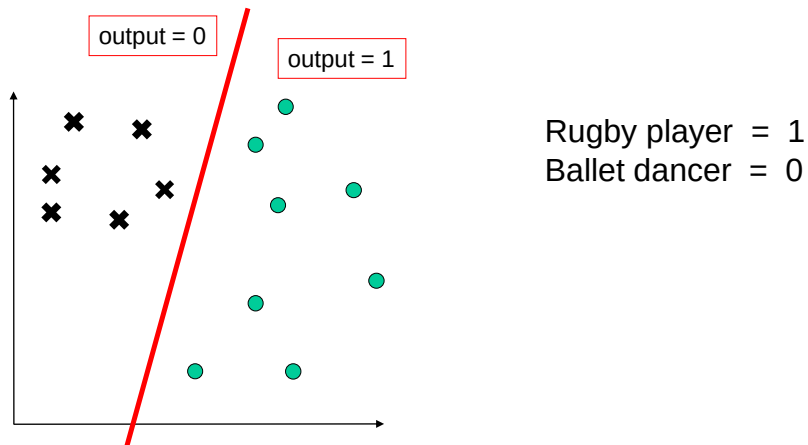
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Perceptron Decision Boundary

$$\text{if } \left(\sum_{i=1}^M x_i w_i \right) > t \quad \text{then } \text{output} = 1, \text{ else } \text{output} = 0$$



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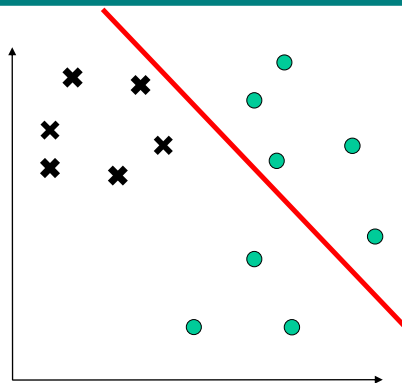
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Perceptron Decision Boundary



Is this a good decision boundary?

$$\text{if } \left(\sum_{i=1}^M x_i w_i \right) > t \quad \text{then } \text{output} = 1, \text{ else } \text{output} = 0$$

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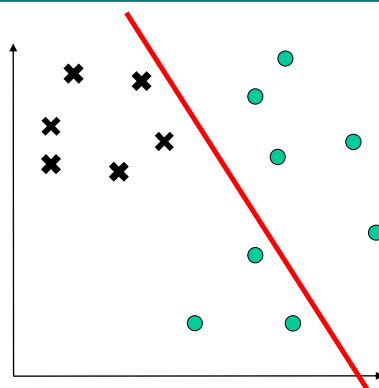
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Perceptron Decision Boundary



$$w_1 = 1.0$$

$$w_2 = 0.2$$

$$t = 0.05$$

$$\text{if } \left(\sum_{i=1}^M x_i w_i \right) > t \quad \text{then output} = 1, \text{ else output} = 0$$

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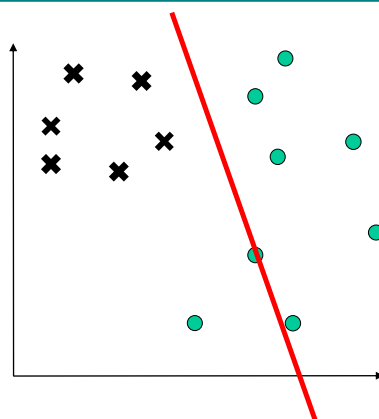
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Perceptron Decision Boundary



$$w_1 = 2.1$$

$$w_2 = 0.2$$

$$t = 0.05$$

$$\text{if } \left(\sum_{i=1}^M x_i w_i \right) > t \quad \text{then output} = 1, \text{ else output} = 0$$

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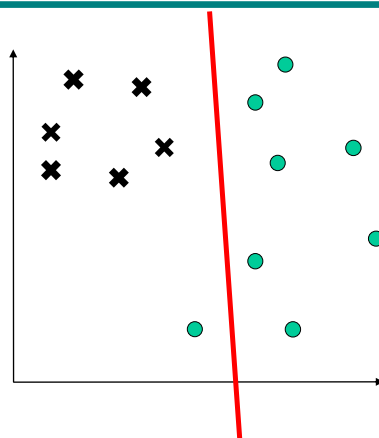
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Perceptron Decision Boundary



$$w_1 = 1.9$$

$$w_2 = 0.02$$

$$t = 0.05$$

$$\text{if } \left(\sum_{i=1}^M x_i w_i \right) > t \quad \text{then output} = 1, \text{ else output} = 0$$

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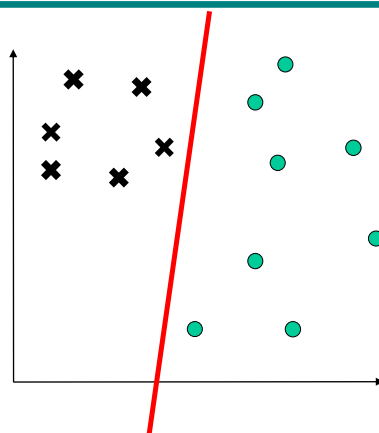
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Perceptron Decision Boundary



$$w_1 = -0.8$$

$$w_2 = 0.03$$

$$t = 0.05$$

Changing the weights/threshold makes the decision boundary move.

Pointless / impossible to do it by hand – only ok for simple 2-D case.

We need an algorithm....

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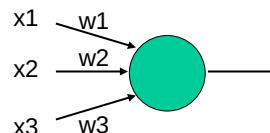
Calculations

$$x = [1.0, 0.5, 2.0]$$

$$w = [0.2, 0.5, 0.5]$$

$$t = 1.0$$

$$a = \sum_{i=1}^M x_i w_i$$



Q1. What is the activation, a , of the neuron?

Q2. Does the neuron fire?

Q3. What if we set threshold at 0.5 and weight #3 to zero?

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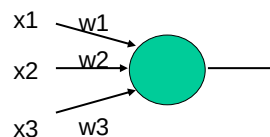
Calculations

$$x = [1.0, 0.5, 2.0]$$

$$w = [0.2, 0.5, 0.5]$$

$$t = 1.0$$

$$a = \sum_{i=1}^M x_i w_i$$



Q1. What is the activation, a , of the neuron?

$$a = \sum_{i=1}^M x_i w_i = (1.0 \times 0.2) + (0.5 \times 0.5) + (2.0 \times 0.5) = 1.45$$

Q2. Does the neuron fire?

if (activation > threshold) output=1 else output=0

.... So yes, it fires.

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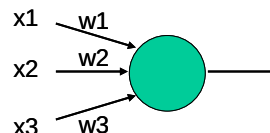
Calculations

$$x = [1.0, 0.5, 2.0]$$

$$w = [0.2, 0.5, 0]$$

$$t = 0.5$$

$$a = \sum_{i=1}^M x_i w_i$$



Q3. What if we set threshold at 0.5 and weight #3 to zero?

$$a = \sum_{i=1}^M x_i w_i = (1.0 \times 0.2) + (0.5 \times 0.5) + (2.0 \times 0.0) = 0.45$$

if (activation > threshold) output=1 else output=0

.... So no, it does not fire..

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Need a More Sophisticated Model

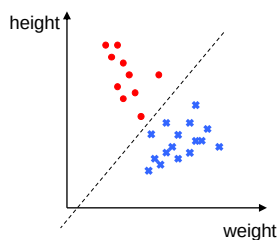
if (weight > t) then "player" else "dancer"



if ($f(\vec{x}) > t$) then "player" else "dancer"

$x_1 = \text{height (cm)}$

$x_2 = \text{weight (kg)}$



The Perceptron

$$f(\vec{x}) = (w_1 * x_1) + (w_2 * x_2)$$

$$= \sum_{i=1}^d w_i x_i$$

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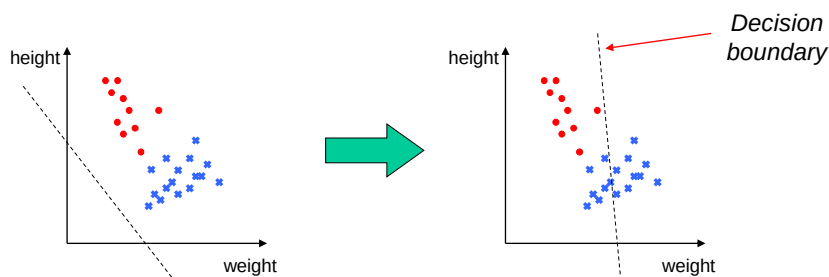
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The Perceptron

if $f(\vec{x}) > t$ then "player" else "dancer"

$$f(\vec{x}) = (w_1 * x_1) + (w_2 * x_2)$$

$$= \sum_{i=1}^d w_i x_i$$



w_1 , w_2 and t change the position of the DECISION BOUNDARY

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The Perceptron

Model

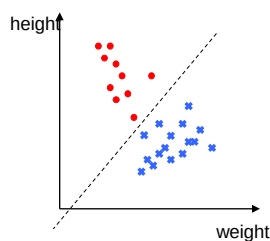
if $\sum_{i=1}^d w_i x_i > t$ then $\hat{y} = 1$ else $\hat{y} = 0$ $\begin{cases} \text{"player"} = 1 \\ \text{"dancer"} = 0 \end{cases}$

Error function

Number of mistakes (a.k.a. classification error)

Learning algo.

??? need to optimise the w values...



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Perceptron Learning Rule

$$\text{new weight} = \text{old weight} + \underbrace{0.1 \times (\text{target} - \text{output}) \times \text{input}}_{\text{update}}$$

What weight updates do these cases produce?

if... (target = 0, output = 0) then	update =	?
if... (target = 0, output = 1) then	update =	?
if... (target = 1, output = 0) then	update =	?
if... (target = 1, output = 1) then	update =	?

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Perceptron Learning Algorithm

```

initialise weights to random numbers in range -1 to +1
for n = 1 to NUM_ITERATIONS
    for each training example (x,y)
        calculate activation
        for each weight
            update weight by learning rule
        end
    end
end
end

```

Perceptron convergence theorem:

If the data is linearly separable, then application of the Perceptron learning rule will find a separating decision boundary, within a finite number of iterations

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We can Rearrange the Decision Rule

$$\text{if } \left(\sum_{i=1}^M x_i w_i \right) > t \quad \text{then output} = 1, \text{ else output} = 0$$

Introducing bias

$$\text{if } \left(\sum_{i=1}^M x_i w_i \right) + (x_0 \times w_0) > t \quad \text{then output} = 1, \text{ else output} = 0$$

$$\text{if } \left(\sum_{i=0}^M x_i w_i \right) > t \quad \text{then output} = 1, \text{ else output} = 0$$

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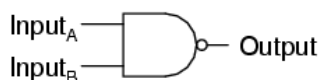
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Example

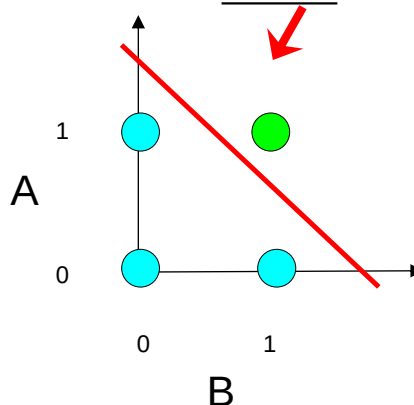
NAND gate



A	B	Output
0	0	1
0	1	1
1	0	1
1	1	0



With a perceptron...
the decision boundary is
LINEAR



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Example

- Inputs: x_0, x_1, x_2 , with input x_0 held constant at 1.
- Threshold: 0.5

Learning rate: 0.1

Input				Output										Error			Correction			Final weights		
Sensor values				Desired output			Initial weights			Per sensor			Sum	Network	Error	Correction	Final weights					
x_0	x_1	x_2		z	w_0	w_1	w_2	c_0	c_1	c_2		s		n	e	d	w_0	w_1	w_2			
								$x_0 * w_0$	$x_1 * w_1$	$x_2 * w_2$		$c_0 + c_1 + c_2$		if $s > t$ then 1, else 0	$z - n$	$r * e$						
1	0	0		1	0	0	0	0	0	0		0		0	1	+0.1	0.1	0	0			
1	0	1		1	0.1	0	0	0.1	0	0		0.1		0	1	+0.1	0.2	0	0.1			
1	1	0		1	0.2	0	0.1	0.2	0	0		0.2		0	1	+0.1	0.3	0.1	0.1			
1	1	1		0	0.3	0.1	0.1	0.3	0.1	0.1		0.5		0	0	0	0.3	0.1	0.1			
1	0	0		1	0.3	0.1	0.1	0.3	0	0		0.3		0	1	+0.1	0.4	0.1	0.1			
1	0	1		1	0.4	0.1	0.1	0.4	0	0.1		0.5		0	1	+0.1	0.5	0.1	0.2			
1	1	0		1	0.5	0.1	0.2	0.5	0.1	0		0.6		1	0	0	0.5	0.1	0.2			
1	1	1		0	0.5	0.1	0.2	0.5	0.1	0.2		0.8		1	-1	-0.1	0.4	0	0.1			
1	0	0		1	0.4	0	0.1	0.4	0	0		0.4		0	1	+0.1	0.5	0	0.1			
1	0	1		1	0.5	0	0.1	0.5	0	0.1		0.6		1	0	0	0.5	0	0.1			
1	1	0		1	0.5	0	0.1	0.5	0	0		0.5		0	1	+0.1	0.6	0.1	0.1			
1	1	1		0	0.6	0.1	0.1	0.6	0.1	0.1		0.8		1	-1	-0.1	0.5	0	0			
1	0	0		1	0.5	0	0	0.5	0	0		0.5		0	1	+0.1	0.6	0	0			
1	0	1		1	0.6	0	0	0.6	0	0		0.6		1	0	0	0.6	0	0			
1	1	0		1	0.6	0	0	0.6	0	0		0.6		1	0	0	0.6	0	0			
1	1	1		0	0.6	0.1	0.1	0.6	0.1	0.1		0.6		1	EME (NUST)	-1	0.5	-0.1	-0.1			

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Example

1	0	0		1	0.5	-0.1	-0.1	0.5	0	0		0.5		0	1	+0.1	0.6	-0.1	-0.1			
1	0	1		1	0.6	-0.1	-0.1	0.6	0	-0.1		0.5		0	1	+0.1	0.7	-0.1	0			
1	1	0		1	0.7	-0.1	0	0.7	-0.1	0		0.6		1	0	0	0.7	-0.1	0			
1	1	1		0	0.7	-0.1	0	0.7	-0.1	0		0.6		1	-1	-0.1	0.6	-0.2	-0.1			
1	0	0		1	0.6	-0.2	-0.1	0.6	0	0		0.6		1	0	0	0.6	-0.2	-0.1			
1	0	1		1	0.6	-0.2	-0.1	0.6	0	-0.1		0.5		0	1	+0.1	0.7	-0.2	0			
1	1	0		1	0.7	-0.2	0	0.7	-0.2	0		0.5		0	1	+0.1	0.8	-0.1	0			
1	1	1		0	0.8	-0.1	0	0.8	-0.1	0		0.7		1	-1	-0.1	0.7	-0.2	-0.1			
1	0	0		1	0.7	-0.2	-0.1	0.7	0	0		0.7		1	0	0	0.7	-0.2	-0.1			
1	0	1		1	0.7	-0.2	-0.1	0.7	0	-0.1		0.6		1	0	0	0.7	-0.2	-0.1			
1	1	0		1	0.7	-0.2	-0.1	0.7	-0.2	0		0.5		0	1	+0.1	0.8	-0.1	-0.1			
1	1	1		0	0.8	-0.1	-0.1	0.8	-0.1	-0.1		0.6		1	-1	-0.1	0.7	-0.2	-0.2			
1	0	0		1	0.7	-0.2	-0.2	0.7	0	0		0.7		1	0	0	0.7	-0.2	-0.2			
1	0	1		1	0.7	-0.2	-0.2	0.7	0	-0.2		0.5		0	1	+0.1	0.8	-0.2	-0.1			
1	1	0		1	0.8	-0.2	-0.1	0.8	-0.2	0		0.6		1	0	0	0.8	-0.2	-0.1			
1	1	1		0	0.8	-0.2	-0.1	0.8	-0.2	-0.1		0.5		0	0	0	0.8	-0.2	-0.1			
1	0	0		1	0.8	-0.2	-0.1	0.8	0	0		0.8		1	0	0	0.8	-0.2	-0.1			
1	0	1		1	0.8	-0.2	-0.1	0.8	0	-0.1		0.7		1	0	0	0.8	-0.2	-0.1			

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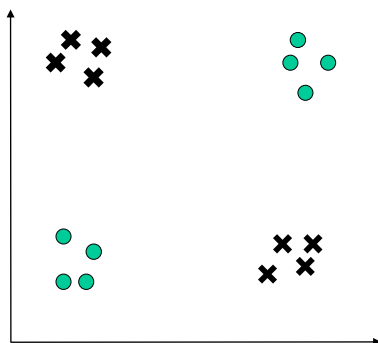
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Can a Perceptron Solve this Problem?



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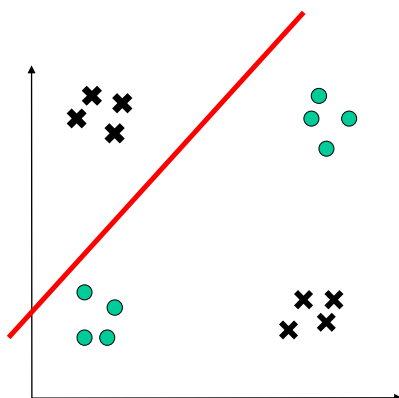
EME (NUST)

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Can a Perceptron Solve this Problem?

No



Perceptrons only solve
LINEARLY SEPARABLE
problems

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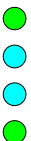
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Perceptron Decision Boundary

Exclusive-OR gate

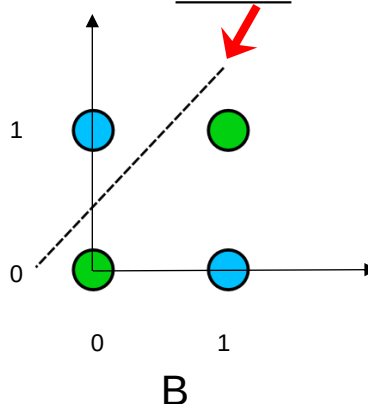


A	B	Output
0	0	0
0	1	1
1	0	1
1	1	0



A

With a perceptron...
the decision boundary is
LINEAR



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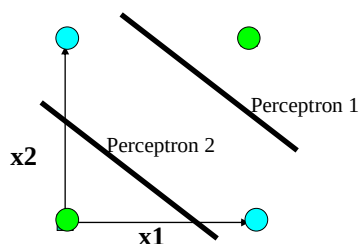
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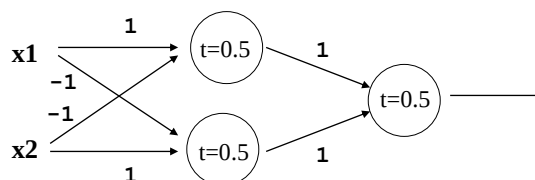
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Multilayer Perceptrons (MLP)



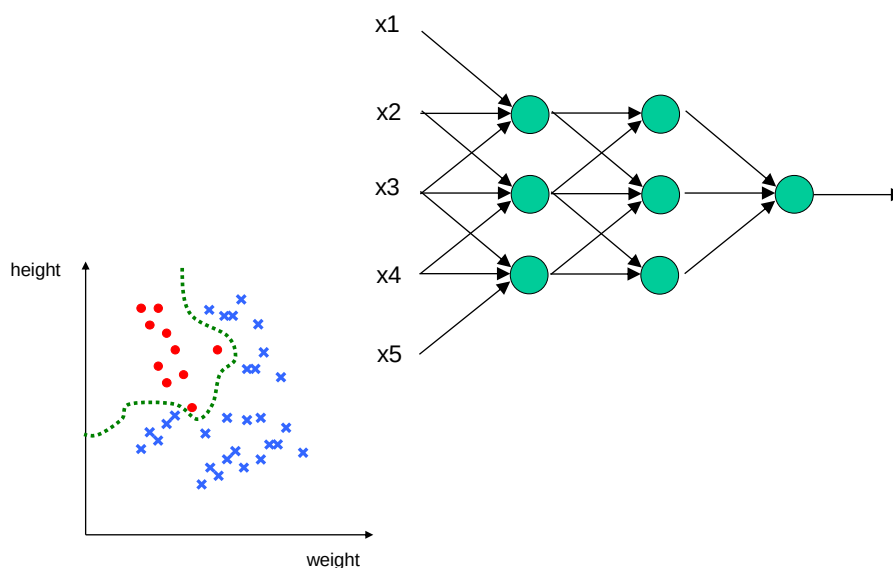
A MultiLayer Perceptron (MLP) can solve XOR and other non-linearly separable problems. The challenge is how to set the weights. The algorithm used is called **BACKPROPAGATION**.

Each node draws one boundary each, then the final node makes a decision using those.



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MLP Decision Boundary – Nonlinear Problems, Solved!



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Neural Networks - Summary

Perceptrons are a (simple) emulation of a neuron.

Layering perceptrons gives you... a multilayer perceptron.

An MLP is one type of neural network – there are others.

An MLP with sigmoid activation functions can solve highly nonlinear problems.

Downside – we cannot use the simple perceptron learning algorithm.

Instead we have the “backpropagation” algorithm.

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Assignment 4

- Submit your project proposal
 - *Title*
 - *Group members (max 4)*
 - *Introduction, scope and aims of your project*
 - *Dataset intend to use*
 - *Methodology to be followed*
 - *Block Diagram*
 - *Proposal should be at least of 2 to 3 pages*
 - *Submission after 1 week*